BIG DATA ANALYTICS





SECTION 1

What is Big Data?

Big Data refers to extremely large and complex datasets that traditional data processing tools cannot effectively handle.
These datasets are generated continuously from various sources, such as social media, websites, mobile applications, sensors, IoT devices, business transactions, scientific experiments, and healthcare systems.

Big Data technologies, such as **Hadoop**, **HDFS**, **MapReduce**, and **Apache Spark**, are designed to store, process, and analyze these large datasets efficiently.
Big Data analytics helps organizations discover patterns, make better decisions, predict future trends, and improve products and services.

5 Vs of Big Data

1.

Volume

Massive amounts of data generated continuously from diverse sources.

2.

Velocity

The rapid speed at which data is created, processed, and analyzed.

3.

Variety

Different forms of data, including structured, unstructured, and semi-structured formats like

4.

Veracity

The reliability, accuracy, and trustworthiness of the data.

5.

Value

The meaningful insights and business benefits derived from analyzing data.

These aspects together create both challenges and opportunities in storing, managing, and leveraging big data for informed decision-making and innovation.

Big Data Applications

Big Data is widely used in many fields to analyze large amounts of information and support better decision-making.

- **Healthcare**: Big Data analyzes patient records, medical images, and genetic information to detect diseases early, predict health risks, and support personalized treatment.
- **Business & Retail**: Companies analyze customer purchases, preferences, and online behavior to understand customer needs, recommend products, and predict future demand.
- **Finance & Banking**: Banks process large volumes of transactions to detect unusual activities, identify fraud, assess financial risks, and support better investment decisions.
- **Social Media**: Platforms analyze millions of posts, comments, and interactions to identify popular trends, understand user interests, and perform sentiment analysis.
- **Transportation**: Data from GPS devices, vehicles, and traffic sensors is used to monitor traffic conditions, predict congestion, optimize routes, and improve transportation services.
- **Manufacturing**: Data collected from machines and production systems helps companies monitor performance, detect equipment problems before failures occur, and improve product quality.
- **Science & Research**: Researchers use Big Data to process massive datasets, such as genomic sequences, climate measurements, and experimental results, helping them discover patterns and make new scientific findings.

What is Hadoop & Why It's Used

Hadoop is an open-source software framework designed for storing and processing large datasets across clusters of commodity hardware.

Why Use Hadoop?

Scalable: Distributes data and processing across many machines, easily handling large data volumes.

Cost-Effective: Runs on affordable commodity hardware, saving costs compared to traditional systems.

Fault-Tolerant: Automatically replicates data to ensure no loss even if a node fails.

Flexible: Supports various data types, including structured, semi-structured, and unstructured data.

Fast Parallel Processing: Uses MapReduce to split tasks and process data in parallel for high performance.

Foundation of Big Data: Core technology behind many big data tools and analytics platforms.

Hadoop enables organizations to efficiently manage and analyze big data that traditional systems cannot handle.

What Is Cloudera and Why Is Linux Used?

Cloudera is a leading enterprise data cloud platform that provides a **comprehensive distribution** of **Apache Hadoop and related big data tools**. It enables organizations to **manage**, **process**, and **analyze vast amounts of data efficiently in a scalable and secure environment.**

Why Use Linux with Cloudera?

- **Stable & High Performance:** Ideal for heavy big data workloads.
- **Open Source Friendly:** Seamless support for Hadoop and related tools.
- **Secure:** Strong protections for sensitive data.
- **Flexible:** Customizable environment for managing clusters.
- **Supported:** Large community and enterprise backing for troubleshooting.

Together, Cloudera on Linux forms a powerful and reliable platform for big data analytics and processing.

Cloudera Distribution

- Hadoop has more than one Distributions
  e.g Cloudera, MAPR ,HotonWorks.

- **Cloudera's** open source platform distribution, which includes Apache Hadoop and is **built specifically to meet enterprise demands**.

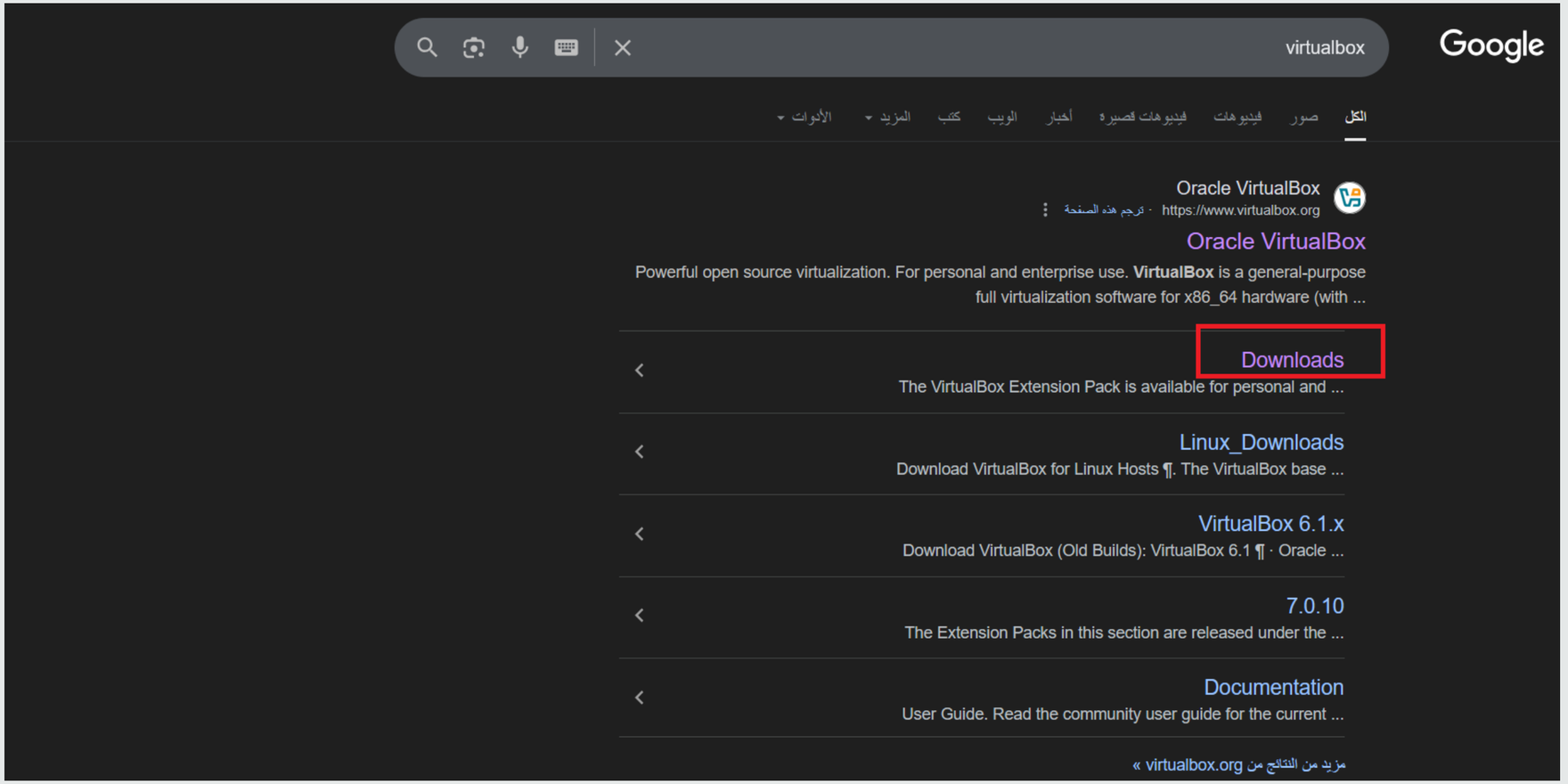
Cloudera Installation

Cloudera QuickStart VM Installation – Prerequisites

- A virtual machine such as Oracle Virtual Box or VMWare
- RAM of 12+ GB. That is 4+ GB for the operating system and 8+ GB for Cloudera
- 80GB hard disk

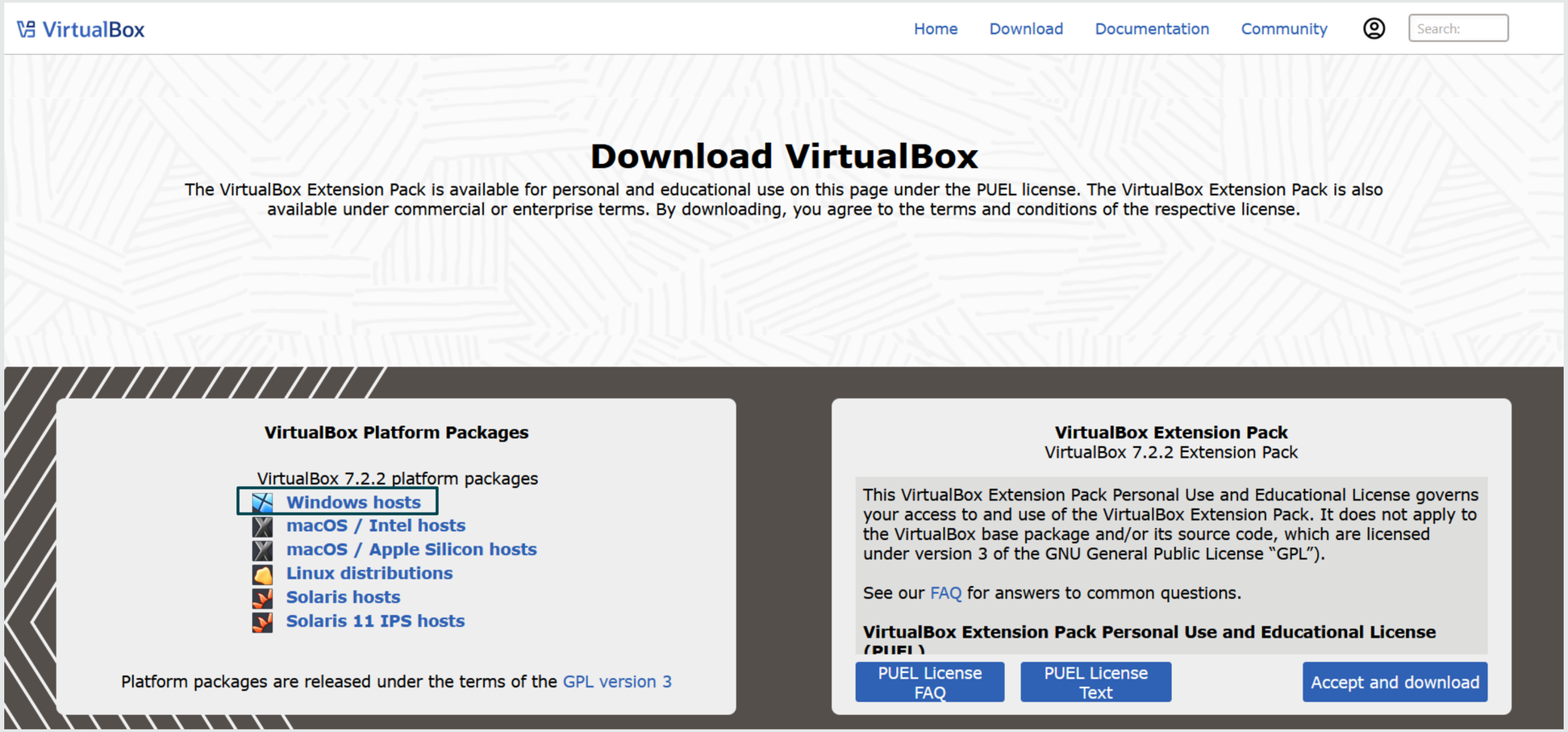
Cloudera installation

1- Download Virtual Box



Cloudera installation

Download Virtual Box: Download the version matches with your OS and install it on your Device. https://www.virtualbox.org/wiki/Downloads



Cloudera installation

2- Download Cloudera from:

https://downloads.cloudera.com/demo_vm/virtualbox/cloudera-quickstart-vm-5.13.0-0-virtualbox.zip

Cloudera installation

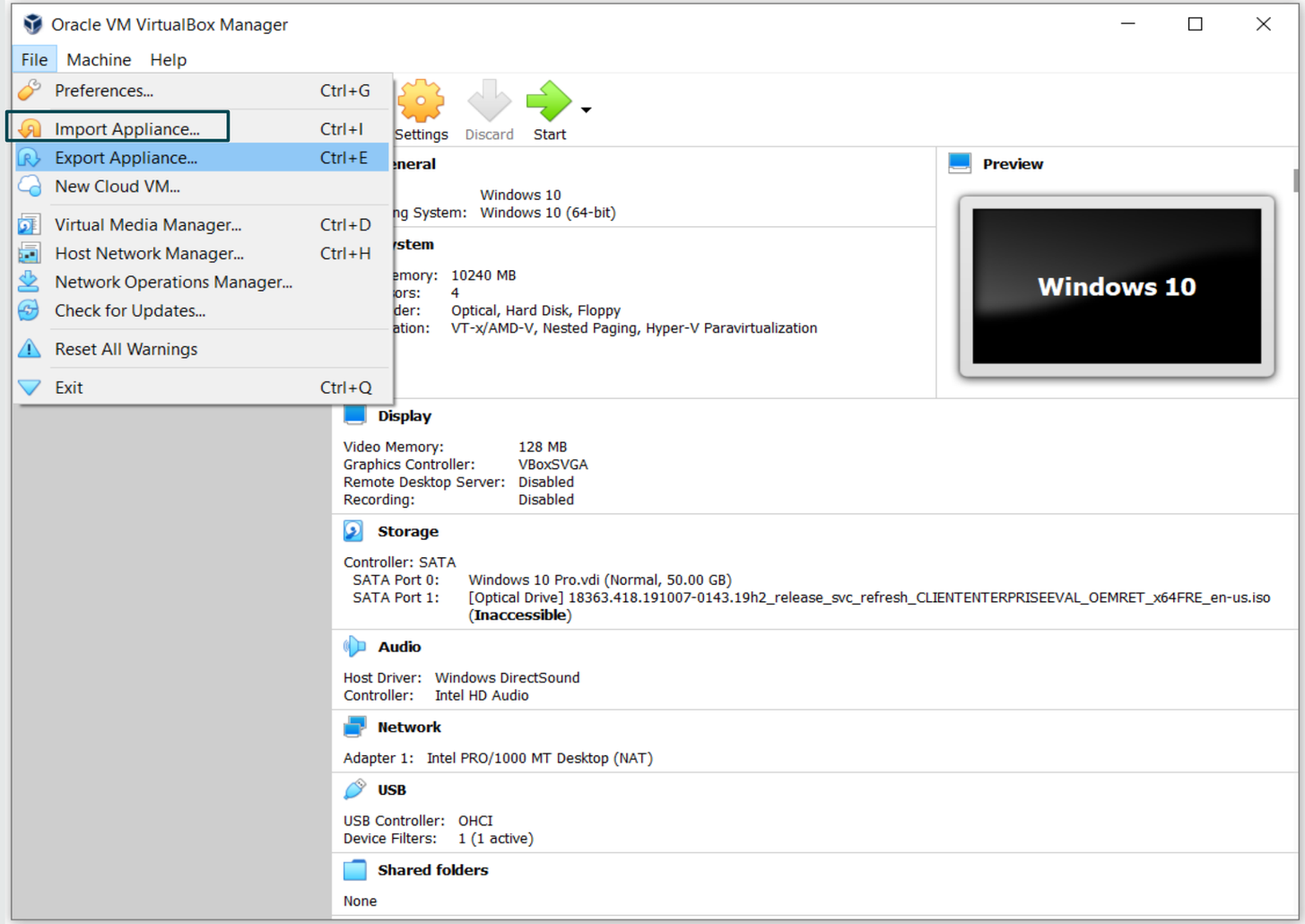
4-
Launch VirtualBox.

b) Click on File - Import appliance

and select the .ovf file from the location
where you have saved the extracted Cloudera Quickstart VM files.

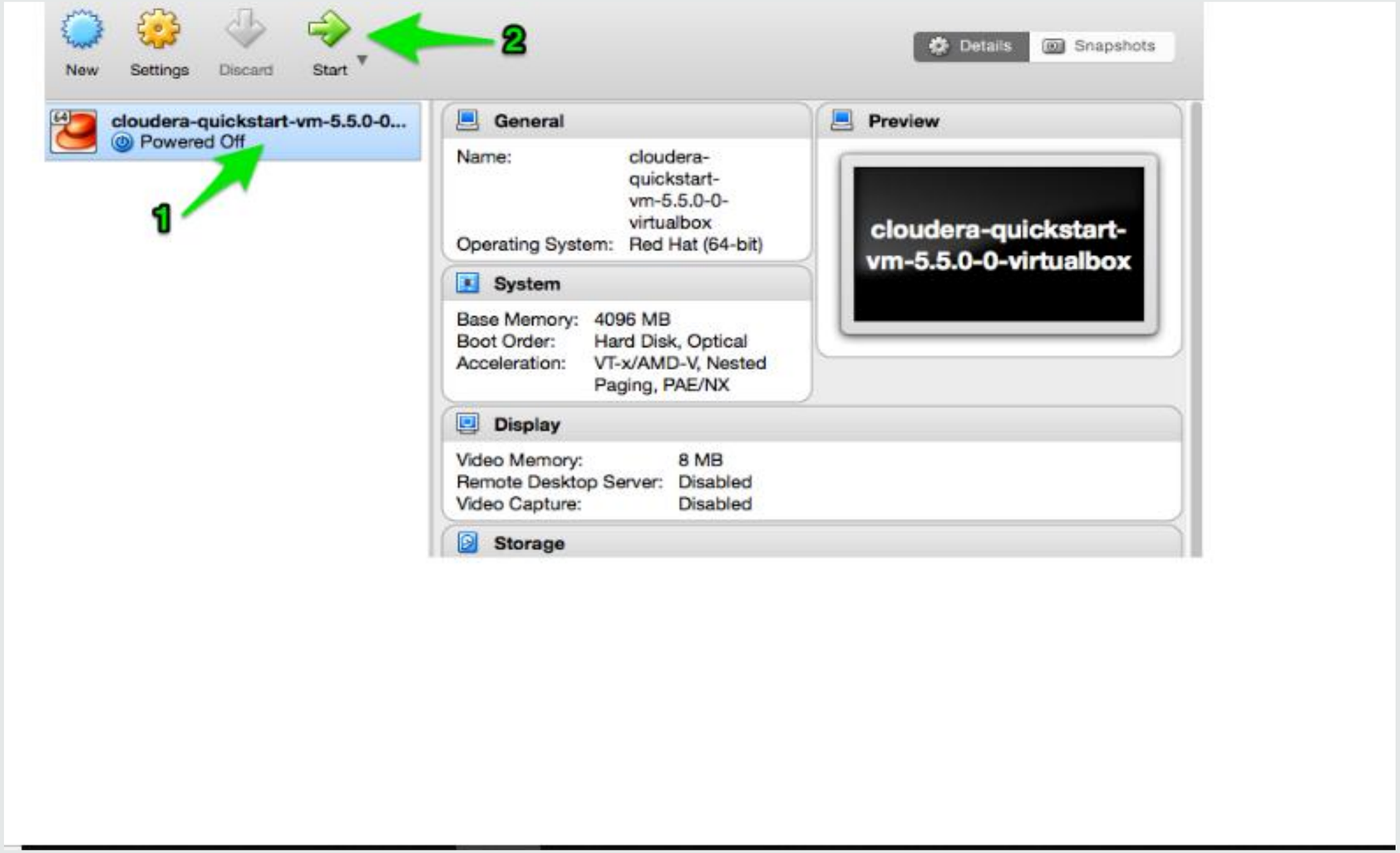
c) Click Next.

d) Click Import.



Cloudera installation

5-
Start your Cloudera Quickstart VM instance.
a) Select cloudera-quickstart-vm from the left panel.
b) Click the Start (green arrow) icon.



Cloudera Installation

Note: The VM can take some time to start up. You'll see the blue Cloudera desktop and the Welcome to Cloudera Quickstart VM page once it's ready.

Open the Terminal.
Select Applications > System Tools > Terminal.

Basic Linux Commands

pwd → Show current working directory

ls → List files and directories

cd <directory> → Change directory

mkdir <name> → Create a new directory

rmdir <name> → Remove an empty directory

clear → Clear the terminal screen

THANK YOU